

AI/ML ENGINEER — 90 DAY JOURNEY

DAY 1 — PYTHON FOUNDATIONS

Progress: Day 1 / 90 | 89 days remaining | Study target: 5 hours

Today's Mission

Build the programming foundation needed for AI/ML. Today you learn variables, data types, operators, input/output, type conversion, and Python literals.

■ MOST IMPORTANT TODAY

Master **variables + data types + literals + input** → **processing** → **output**. Understand what each line of a small Python program does.

5-Hour Study Schedule

Time	Focus	What to learn
Hour 1	Setup + first programs	Python, VS Code, .py files, print(), comments, variables
Hour 2	Variables + data types ■	int, float, str, bool, type(), assigning values
Hour 3	Operators + literals	Arithmetic, comparison, %, //, **, common literal forms
Hour 4	Input + output ■■■	input(), print(), converting input, small programs
Hour 5	Practice	Solve challenges without following a tutorial

1. Variables

A variable is a name that refers to a value. Variables let programs store and reuse information.

```
name = "Jahanur"
age = 22
height = 5.8
is_student = True
```

2. Python Data Types

The four types you must know today: **int** (whole number), **float** (decimal), **str** (text), and **bool** (True/False).

```
print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
```

3. Python Literals ■

A **literal** is a fixed value written directly in Python source code. It represents a value exactly as written, rather than being the name of a variable.

```
age = 22 # 22 is an integer literal
price = 99.50 # 99.50 is a float literal
name = "Jahanur" # "Jahanur" is a string literal
passed = True # True is a Boolean literal
```

```
items = [1, 2, 3] # [1, 2, 3] is a list literal
```

Easy way to remember: the actual value written in the code is the literal. In **age = 22**, **age** is the variable and **22** is the literal.

4. Operators

Arithmetic: +, -, *, /, %, **, //

Comparison: ==, !=, >, <, >=, <=

```
a = 10
b = 3
print(a + b)
print(a % b)    # remainder
print(a ** b)   # power
print(a // b)   # floor division
```

5. Type Conversion

Input normally arrives as text. Convert it when you need a number.

```
age = input("Enter age: ")
age = int(age)
print(age + 1)
```

6. Input → Processing → Output ■■■■

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))

print("Hello", name)
print("Next year you will be", age + 1)
```

7. Practice Challenges

1. Ask for name, age, and city and print an introduction.
2. Ask for two numbers and calculate +, -, *, /, and %.
3. Ask for three subject marks and calculate total and average.
4. Convert Celsius to Fahrenheit: $F = (C \times 9/5) + 32$.
5. Create a shopping bill using price × quantity.

8. Official Documentation

Python Tutorial — Introduction: <https://docs.python.org/3/tutorial/introduction.html>

Python Documentation: <https://docs.python.org/3/>

Focus on numbers, text, variables, expressions, `print()`, and basic programming concepts.

9. Day 1 Self-Test — Questions + Answers

First try to answer each question yourself. Then compare your answer with the explanation.

1. What is Python?

Answer: Python is a high-level, general-purpose programming language. It is widely used for web development, automation, data analysis, AI, and machine learning.

2. What is a variable?

Answer: A variable is a named reference used to store or refer to a value in a program. Example: **age = 22**.

3. What is a data type?

Answer: A data type describes the kind of value stored or represented, such as an integer, decimal number, text, or Boolean.

4. Difference between int and float?

Answer: **int** represents whole numbers, such as 10 or -3. **float** represents numbers with a decimal part, such as 10.5 or -3.2.

5. What is a string?

Answer: A string is a sequence of text characters. In Python, strings are usually written inside quotes. Example: **name = "Jahanur"**.

6. What is a Boolean?

Answer: A Boolean is a data type with two values: **True** or **False**. It is commonly used for conditions and logical decisions.

7. What does `input()` do?

Answer: `input()` reads information entered by the user and returns it as a string. Example: **name = input("Enter name: ")**.

8. What does `print()` do?

Answer: `print()` displays information in the program's output/console. Example: **print("Hello")**.

9. Why use `int(input())`?

Answer: Because `input()` returns text. `int(input())` converts the entered text into an integer so you can perform numerical operations.

10. Difference between `/` and `//`?

Answer: `/` performs normal division. `//` performs floor division, giving the quotient rounded down to the nearest whole number for integer operands.

11. What does `%` do?

Answer: The `%` operator returns the remainder after division. Example: **10 % 3** gives **1**.

12. What does ****** do?

Answer: The ****** operator performs exponentiation (power). Example: **2 ** 3** gives **8**.

13. What does **type()** do?

Answer: **type()** tells you the data type/class of a value. Example: **type(22)** returns **int**.

14. What are literals? Give examples.

Answer: A **literal** is a fixed value written directly in Python source code. Examples: **22** is an integer literal, **99.5** is a float literal, **"Jahanur"** is a string literal, and **True** is a Boolean literal. In **age = 22**, **age** is the variable and **22** is the literal.

10. Day 1 Completion Checklist

- Watched the assigned lesson section
- Practiced variables and data types
- Understood literals and their examples
- Practiced operators and type conversion
- Built the five challenges
- Read the selected official documentation
- Reviewed all 14 questions and answers

■ MOTIVATION

Never give up Jahanur, I can do it.

You do not need to become an AI Engineer today. You only need to win Day 1. One day at a time. 5 hours × 90 days = 450 hours of focused progress.